



01 What It Is

Gradient is a pitch-shifting delay with two independent units, A and B. Each one shifts pitch and delays its own signal, and can run alone, locked to the other by a fixed interval, or feeding into the other's feedback path.

It's built for pitch-shifted echoes, harmonized delays, and self-oscillating textures rather than one fixed effect. Mode decides how A and B combine; everything else on the panel shapes what each unit does on its own.

02 Signal Path

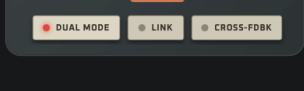
Each unit runs this same chain independently. Mode (see The Panel below) decides how A and B's outputs combine.



03 The Panel

MODE SPLICE PITCH DELAY / REGEN DRIFT / OUTPUT

MODE



Mode Dual Mode / Link / Cross-Feedback

Chooses how units A and B combine. With none selected, unit A runs alone. Dual Mode runs both as independent voices. Link locks B's pitch and delay to A plus a fixed offset. Cross-Feedback routes each unit's output into the other's feedback path.



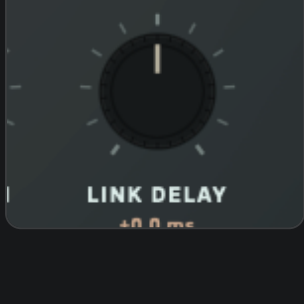
Stereo 0–100% · default 100%

The stereo spread of the delay signal. At 0% it's mono, at 100% it's full stereo width.



Link Pitch -24 to +24 st · default 0 st

Only active when Link is on. Sets B's pitch as a fixed offset from A's, instead of using B's own Pitch knob.



Link Delay -1000 to +1000 ms · default 0 ms

Only active when Link is on. Sets B's delay as a fixed offset from A's, instead of using B's own Delay knob.

SPLICE

Applies separately to units A and B.



Splice Mode Normal / Soft / Smart · default Normal

The pitch-shift algorithm's splicing behavior. Normal is the baseline. Soft and Smart trade off latency, artifacts, and pitch accuracy differently.



Crossfade Length 1–14 ms · default 8 ms

How long each splice's grain overlap lasts. Longer values smooth the splice at the cost of a softer transient.

PITCH

Applies separately to units A and B.



Pitch -24 to +24 st · default 0 st

This unit's pitch shift amount, in semitones.

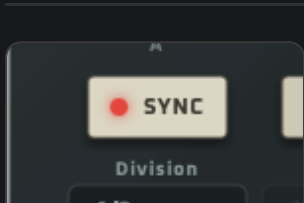


Fine -50 to +50 ct · default 0 ct

Fine pitch trim, in cents, for tuning between semitones.

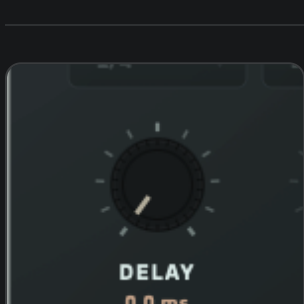
DELAY / REGEN

Applies separately to units A and B.



Delay Sync on/off · default off

Locks this unit's delay time to a tempo-synced note division instead of the Delay knob's free-running milliseconds. The Division dropdown picks the note value once Sync is on.



Delay 0–1000 ms · default 0 ms

This unit's delay time. Unused while Delay Sync is on.

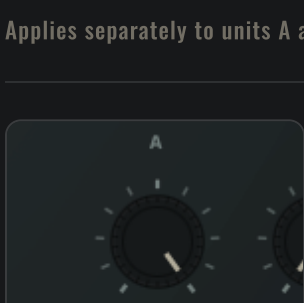


Regen 0–350% · default 0%

Regeneration amount for this unit's delay. The range runs well past unity and self-oscillates at the high end.

DRIFT / OUTPUT

Applies separately to units A and B.



Drift 0–100% · default 0%

Adds slow pitch and time wander to this unit.



Mix 0–100% · default 50%

This unit's own dry/processed balance.



Out -24 to +12 dB · default 0 dB

This unit's output trim.

A Bypass switch sits at the top right of the panel. It mutes the effect entirely, passing dry signal only, for quick A/B comparison.